

# DINESH P U

## SOFTWARE DEVELOPER

Coimbatore, Tamil Nadu, India | +91 8072814839 | [dineshkumar46247@gmail.com](mailto:dineshkumar46247@gmail.com)

### EXECUTIVE SUMMARY

Versatile Software Developer with over 4 years of hands-on experience in full-lifecycle software development, application maintenance, production engineering, and database management. Specialized in high-consequence enterprise applications within the **aviation software domain**, supporting world-class airlines and aerospace organizations. Proven track record in conducting complex root-cause analysis, implementing critical business logic enhancements, optimizing database interactions, and ensuring uninterrupted operational stability.

Recognized for high adaptability and early adoption of modern **AI-assisted software development methodologies**. Highly proficient in leveraging OpenAI Codex and structured AI prompt engineering to accelerate code understanding, debug intricate production issues, streamline test-case generation, and deliver robust software features on tight timelines.

### TECHNICAL EXPERTISE

|                        |                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------|
| Core Programming:      | Java, Python, SQL, PL/SQL, T-SQL                                                                                 |
| Database Systems:      | Oracle Database, MariaDB, Microsoft SQL Server                                                                   |
| Frameworks & UI:       | Java Swing, HTML5, Relational Database Architectures                                                             |
| Developer Tools:       | DBeaver, Heidi SQL, SQL Server Management Studio (SSMS), Squirrel SQL Client, OpenAI Codex                       |
| Engineering Practices: | Root Cause Analysis (RCA), Debugging, Requirement Analysis, Unit Testing, Regression Testing, Production Support |
| AI & Automation:       | AI-Assisted Software Engineering, Prompt Engineering, Automated Code Tracing, AI Debugging & Validation          |
| Soft Skills:           | Problem Solving, Cross-Functional Team Collaboration, Adaptability, Technical Communication                      |

### PROFESSIONAL EXPERIENCE

|                                                                                                                                                         |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Software Engineer — Ultramain Software India Pvt. Ltd.                                                                                                  | April 2022 – Aug 2026 |
| Key Aviation Client Engagements: Emirates   Cathay Pacific Airways   AISG   Virgin Galactic   Virgin Atlantic Airways   Starlux Airlines   SRT Malaysia |                       |

Played a pivotal role in the engineering and support of complex enterprise aviation software platforms designed to optimize asset management, regulatory compliance, operational workflows, and maintenance execution.

### **Core Engineering & Application Maintenance:**

- Developed, tested, and deployed business-critical application enhancements utilizing Java, Java Swing, HTML, and SQL.
- Maintained multi-database environments (Oracle, MariaDB, MS SQL Server), executing safe production data corrections, script execution, and operational database support.
- Leveraged industry-standard SQL clients including DBeaver, SSMS, Heidi SQL, and Squirrel SQL to inspect complex schemas, optimize queries, and troubleshoot data-level discrepancies.
- Investigated complex application bugs by systematically tracing multi-layered Java code flow, database triggers, and core business logic to deliver permanent fixes.
- Analyzed end-user change requests, translated business needs into concrete technical specifications, and conducted thorough manual unit and regression testing.

### **AI-Assisted Development & Productivity Innovations:**

- Integrated OpenAI Codex and advanced AI prompting techniques into daily engineering workflows to drastically accelerate issue diagnosis, source code navigation, and feature delivery.
- Formulated structured, context-rich technical prompts to articulate complex business logic requirements and edge cases to AI tools, obtaining precise code snippets and structural insights.
- Applied AI assistance to rapidly comprehend legacy codebase structures, generate comprehensive test cases, automate repetitive boilerplate code generation, and formulate effective debugging strategies.
- Maintained complete oversight and accountability for overall code quality by systematically reviewing, refactoring, and validating all AI-generated suggestions against strict aviation domain requirements before implementation.

### **Key Technical Responsibilities Held:**

- End-to-end requirement analysis, change request estimation, and system design alignment.
- Design, development, and unit testing of bespoke application features and module enhancements.
- Production issue escalation handling, root-cause investigation, and deployment of emergency patches.
- Writing, documenting, and executing manual test suites to safeguard against regression defects.
- Direct collaboration with cross-functional support, maintenance, and technical account teams to meet customer SLAs.

## **HONORS & AWARDS**

### **□ Ultra Star Employee Award**

Conferred by **Ultramain Software India Pvt. Ltd.** in recognition of outstanding performance, exceptional dedication, and consistently completing a high volume of complex technical assignments within strict project timelines without compromising quality.

## PROJECTS & CERTIFICATIONS

---

### Technical Projects:

- **Face Recognition Application (Python) – GUVI:** Designed and implemented a computer vision face recognition tool using Python programming principles and open-source libraries.
- **Green Stock (Web Development):** Developed a responsive web-based platform as part of comprehensive full-stack web development training.

### Certifications & Specialized Training:

|                                     |                                     |
|-------------------------------------|-------------------------------------|
| <b>Python Face Recognition:</b>     | GUVI Professional Certification     |
| <b>Digital Marketing:</b>           | Google Certification                |
| <b>Machine Learning:</b>            | AWS Training Certification          |
| <b>Ethical Hacking:</b>             | Elysium Academy Training            |
| <b>Python Bootcamp:</b>             | Indian Servers Development Bootcamp |
| <b>IoT, AI &amp; ML Internship:</b> | Quantanics Tech Serv Pvt. Ltd.      |

## EDUCATION & PERSONAL DETAILS

---

|                               |                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------|
| <b>B.E. Computer Science:</b> | Nehru Institute of Engineering and Technology, Anna University<br>(2018 – 2022)   CGPA: 7.89 |
| <b>HSC &amp; SSLC:</b>        | St. Michael's Higher Secondary School — HSC (2018): 74.4%  <br>SSLC (2016): 82.8%            |
| <b>Languages Known:</b>       | Tamil (Native), English (Professional Working Proficiency), Kannada<br>(Working Proficiency) |